

**ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY**  
*Department of Software Engineering*

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DATABASE GROUP ASSIGNMENT

**HOTEL MANAGEMENT SYSTEM**  
*Using MySQL and MongoDB*

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# Chapter 1: Introduction

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## 1.1 Background of the Organization

Addis Grand Hotel is a mid-sized three-star hotel located in Addis Ababa, Ethiopia. The hotel operates approximately 120 rooms categorized into four types: Single, Double, Deluxe, and Suite. It provides a full range of hospitality services including accommodation, room service, and restaurant facilities.

Currently, the hotel relies entirely on manual record-keeping and spreadsheets to manage its core operations — bookings, guest information, room status, and payment processing. As the hotel grows, this approach has become increasingly inadequate, prompting the need for a modern database-driven management system.

## 1.2 Problem Statement

The existing manual system at Addis Grand Hotel suffers from several critical limitations that directly impact operational efficiency and customer satisfaction:

- Frequent double bookings due to lack of real-time room availability tracking
- Manual errors in billing and payment records leading to financial discrepancies
- Loss or misplacement of guest records and historical data
- Inefficient handling of room service orders and guest feedback
- Difficulty in generating timely management reports for decision-making
- Increased staff workload and reduced service quality

These challenges collectively result in poor customer experience, revenue leakage, and operational inefficiency. A modern hybrid database system is essential to address these issues.

## 1.3 Objectives

### General Objective

To design and implement a hybrid database system using MySQL (relational) and MongoDB (NoSQL) to automate and improve the overall efficiency of hotel management operations at Addis Grand Hotel.

### Specific Objectives

- To analyze the hotel's data requirements and design a fully normalized relational database schema using MySQL

- To leverage MongoDB for storing flexible and semi-structured data such as guest reviews, feedback, and room service orders
- To enable real-time room availability checking and accurate booking management
- To automate payment tracking and generate management reports (occupancy rate, revenue, guest history)
- To improve data integrity, reduce manual errors, and enforce referential consistency

## 1.4 Scope and Limitations

The system is designed to cover the core operations of Addis Grand Hotel, specifically:

- Room type management and room availability tracking
- Guest registration and profile management
- Booking creation, modification, and check-in/check-out processing
- Payment recording and receipt generation
- Room service order management (via MongoDB)
- Guest reviews and feedback collection (via MongoDB) The following are explicitly out of scope for this phase:
- Online payment gateway integration (e.g., PayPal, Stripe)
- Mobile application development
- Integration with third-party booking platforms (e.g., Booking.com) • Multi-branch or multi-hotel support

## 1.5 Benefits of the System

| Benefit Area           | Description                                                                  |
|------------------------|------------------------------------------------------------------------------|
| No Double Bookings     | Real-time room availability eliminates reservation conflicts                 |
| Faster Check-in/out    | Automated guest lookup and billing reduces processing time                   |
| Flexible Data Handling | MongoDB's schema handles varied guest reviews and service orders efficiently |
| Financial Accuracy     | Automated payment tracking removes billing errors and revenue leakage        |
| Better Reporting       | Instant generation of occupancy, revenue, and guest history reports          |
| Improved Experience    | Faster service and accurate records lead to higher guest satisfaction        |

## 1.6 Methodology and Tools

The project followed the Waterfall software development methodology, progressing through requirements analysis, system design, implementation, testing, and documentation phases in sequence.

### Tools and Technologies Used

| Tool            | Purpose                                 | Category            |
|-----------------|-----------------------------------------|---------------------|
| MySQL 8.0       | Structured transactional data storage   | Relational Database |
| MongoDB         | Document-oriented & semistructured data | NoSQL Database      |
| MySQL Workbench | Schema design and query execution       | IDE / GUI           |
| MongoDB Compass | Collection management and query testing | IDE / GUI           |
| draw.io         | ER Diagram design                       | Design Tool         |
| GitHub          | Version control and collaboration       | DevOps              |

## Chapter 2: Database Design

### 2.1 Entities and Relationships

The following core entities were identified through requirements analysis with hotel staff:

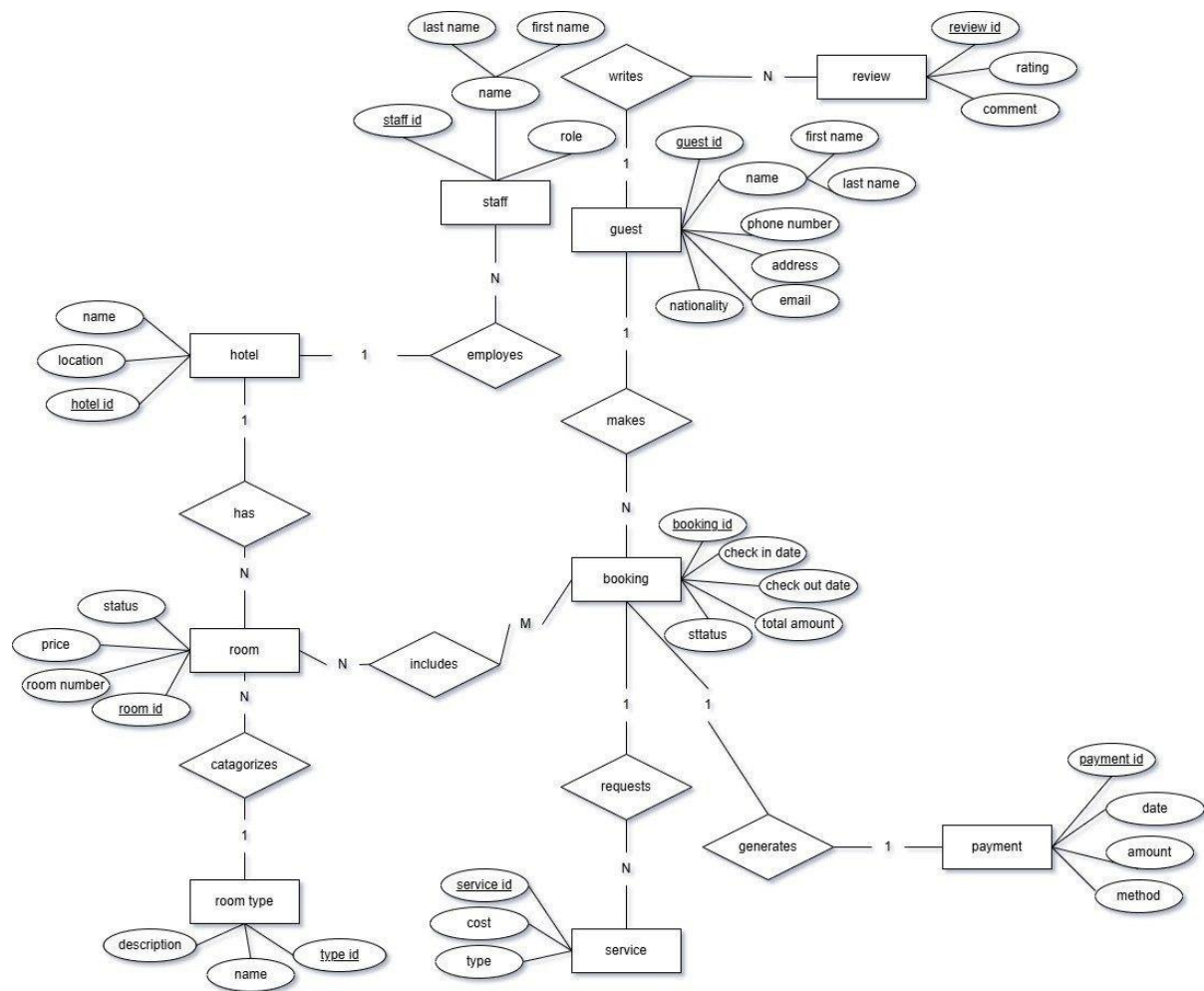
| Entity           | Database | Description                                                                                                                        |
|------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| RoomType         | MySQL    | Categorizes rooms (Single, Double, Deluxe, Suite) with pricing                                                                     |
| Room             | MySQL    | Individual room records with floor number and availability status                                                                  |
| Guest            | MySQL    | Guest personal data, contact information, and ID details                                                                           |
| Booking          | MySQL    | Reservation records linking guests to rooms with dates and status                                                                  |
| BookingRoom      | MySQL    | Junction table resolving M:M relationship between Booking and Room                                                                 |
| Payment          | MySQL    | Payment records linked to bookings with method and amount                                                                          |
| Review           | MongoDB  | Flexible guest reviews with ratings, comments, and tags                                                                            |
| RoomServiceOrder | MongoDB  | Room service orders with variable item arrays and status                                                                           |
| Hotel            | MySQL    | Stores core hotel information including name and location; serves as the top-level entity that rooms and staff are linked to       |
| Staff            | MySQL    | Records hotel employee details including full name, position, contact information, and hire date, linked to the hotel they work at |
| Review           | MySQL    | Stores structured guest review records with a 1–5 rating and comment, linked to both the guest and their booking                   |
| Service          | MySQL    | Records additional services requested during a booking such as room service or airport pickup, with service type and cost          |

## Key Relationships

- Guest → Booking: One-to-Many (1:M) — A guest can make multiple bookings
- Booking ↔ Room: Many-to-Many (M:M) — Resolved via the BookingRoom junction table
- Room → RoomType: Many-to-One (M:1) — Each room belongs to exactly one room type
- Booking → Payment: One-to-Many (1:M) — A booking may have multiple payment installments
- Guest → Review (MongoDB): One-to-Many — Guest IDs are referenced in MongoDB documents
- Booking → RoomServiceOrder (MongoDB): One-to-Many — Orders reference booking\_id
- **Hotel** → **Room**: One-to-Many (1:M) — A hotel contains multiple rooms, each room belongs to one hotel
- **Hotel** → **Staff**: One-to-Many (1:M) — A hotel employs multiple staff members, each staff member belongs to one hotel
- **Guest** → **Review (MySQL)**: One-to-Many (1:M) — A guest can leave multiple reviews, each review is linked to one guest
- **Booking** → **Review**: One-to-Many (1:M) — A booking can have multiple reviews linked to it
- **Booking** → **Service**: One-to-Many (1:M) — A booking can include multiple additional services such as room service or airport pickup

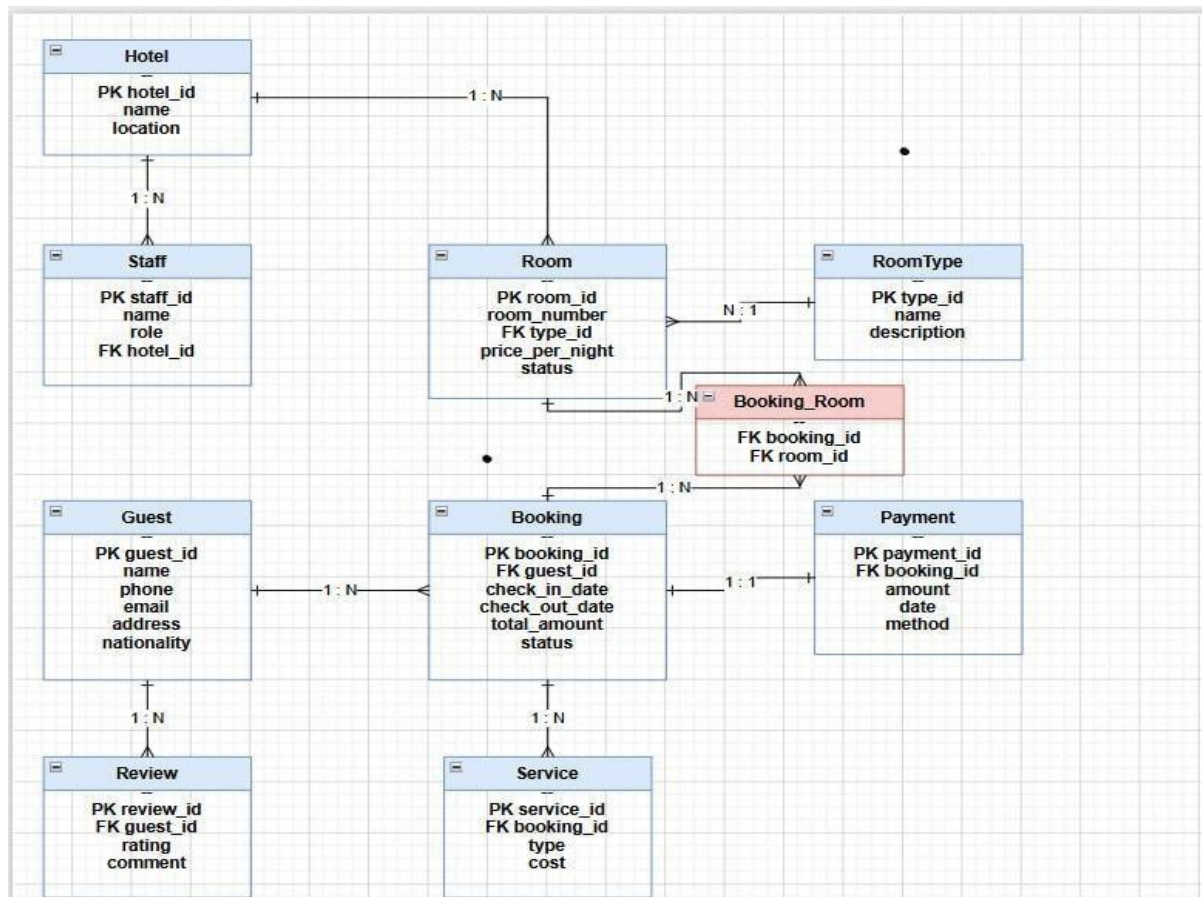
## 2.2 ER Diagram

The Entity-Relationship (ER) diagram below illustrates all entities, their attributes, primary/foreign keys, and the cardinality of each relationship. The diagram was created using draw.io.



## 2.3 Logical Schema

The relational schema was implemented in MySQL with the following table structures:



## 2.4 Normalization (up to BCNF)

The database was normalized through all normal forms up to Boyce-Codd Normal Form (BCNF) to eliminate redundancy and enforce data integrity.

| Form | Requirement                               | Action Taken                                                              |
|------|-------------------------------------------|---------------------------------------------------------------------------|
| 1NF  | Atomic values; no repeating groups        | All attributes hold single values; no multi-valued fields                 |
| 2NF  | No partial dependencies on composite keys | Extracted RoomType data from Room; created BookingRoom junction table     |
| 3NF  | No transitive dependencies                | price_per_night moved to RoomType (not Room); removed derived attributes  |
| BCNF | Every determinant is a candidate key      | Verified all functional dependencies; all determinants are candidate keys |

## 2.5 Input Forms and Output Reports

### Input Forms

The following data entry forms support hotel operations:

- Guest Registration Form — Captures personal details and ID information
- Room Booking Form — Records reservation dates, room type, and special requests
- Check-in / Check-out Form — Manages room assignment and departure
- Room Service Order Form — Records items ordered, quantities, and pricing •
- Payment Receipt Form — Confirms payment amount, method, and date

### Key Output Reports

- Room Availability Report — Real-time view of vacant and occupied rooms
- Daily Revenue Report — Total revenue by day, broken down by room type
- Guest Booking History — Full reservation and payment history per guest
- Occupancy Rate Report — Percentage utilization over a given date range
- Average Review Score Report — Aggregated ratings from MongoDB reviews

## Chapter 3: Implementation

### 3.1 MySQL Implementation

MySQL was selected for all structured, transactional data due to its ACID compliance, robust JOIN capabilities, and support for referential integrity through foreign key constraints.

#### Schema Highlights

- All tables use AUTO\_INCREMENT primary keys for consistent record identification
- Foreign key constraints enforce referential integrity across all relationships
- CHECK constraints validate booking dates (check\_out\_date > check\_in\_date)
- ENUM types are used for room status ('available', 'occupied', 'maintenance') and payment method
- Indexes are applied on frequently queried columns (guest\_id, check\_in\_date)

#### Sample SQL Queries Implemented

| Query Purpose                            | SQL Operation Used                                        |
|------------------------------------------|-----------------------------------------------------------|
| Check room availability for a date range | SELECT with NOT EXISTS subquery on BookingRoom            |
| Calculate total revenue per room type    | JOIN + GROUP BY + SUM(amount) across Payment and RoomType |
| Get guest booking history                | INNER JOIN across Guest, Booking, BookingRoom, Room       |

|                                |                                                   |
|--------------------------------|---------------------------------------------------|
| Compute monthly occupancy rate | Aggregation with DATEDIFF and COUNT               |
| Find top-spending guests       | GROUP BY guest_id ORDER BY SUM(total_amount) DESC |

Full schema: *mysql/schema.sql* | Sample queries: *mysql/queries.sql*

### 3.2 MongoDB Implementation

MongoDB was chosen to manage semi-structured and variable data that does not fit cleanly into a relational schema. Its document model is ideal for guest reviews (with arbitrary tag arrays) and room service orders (with variable item lists).

#### Collections

- reviews — Stores guest\_id, booking\_id, rating (1–5), comment text, and an array of tags
- roomServiceOrders — Stores booking\_id, room\_number, an items array (name, qty, price), total\_cost, and status
- guestFeedback — Stores post-stay feedback with category ratings and open-ended responses

#### Sample MongoDB Operations

| Operation                              | Method Used                                                        |
|----------------------------------------|--------------------------------------------------------------------|
| Insert a guest review                  | db.reviews.insertOne({ guest_id, rating, comment, tags })          |
| Find all reviews for a guest           | db.reviews.find({ guest_id: ObjectId('...') })                     |
| Calculate average rating per room type | db.reviews.aggregate([\$group, \$avg])                             |
| Get all orders for a booking           | db.roomServiceOrders.find({ booking_id: '...' })                   |
| Update order status to 'delivered'     | db.roomServiceOrders.updateOne({ \$set: { status: 'delivered' } }) |

Sample documents: *mongodb/collections.json* | Sample queries: *mongodb/queries.js*

### 3.3 Testing and Results

Both databases were populated with realistic sample data and tested against all defined query requirements. Results confirmed the system performs as expected.

| Test Case                                  | Result  | Notes                                                |
|--------------------------------------------|---------|------------------------------------------------------|
| Book a room for existing guest             | PASS    | Booking created; room status updated to 'occupied'   |
| Attempt to book an occupied room           | PASS    | System correctly rejects double booking              |
| Check-out and mark room available          | PASS    | Room status returns to 'available' on check-out date |
| Generate monthly revenue report            | PASS    | Accurate SUM across all confirmed bookings           |
| Insert guest review in MongoDB             | PASS    | Document stored with tags array intact               |
| Aggregate average ratings                  | PASS    | Correct average computed per room type via pipeline  |
| Cross-reference booking_id (MySQL→MongoDB) | PARTIAL | Manual ID sync required; no automatic trigger yet    |

### 3.4 Challenges, Bugs & Improvements

#### Challenges Encountered

- Data synchronization between MySQL and MongoDB: guest\_id and booking\_id must be manually kept consistent across both systems, as there is no native crossdatabase referential integrity
- Schema design for normalization: determining the correct level of decomposition (especially for BCNF) required multiple iterations
- MySQL date handling: initial booking queries used incompatible date format strings, requiring format standardization

#### Bugs Resolved

- Date format mismatch in booking availability query — fixed by standardizing all date inputs to YYYY-MM-DD format
- Missing NOT NULL constraints on booking dates — added to prevent incomplete booking records
- MongoDB aggregation pipeline returning incorrect room averages — fixed by correctly grouping on room\_type field

#### Recommended Future Improvements

1. Implement user authentication with role-based access control (admin, receptionist, manager)

2. Build a backend integration layer (Node.js or PHP) to orchestrate operations across both databases
3. Add real-time notifications for new bookings and check-out reminders
4. Introduce automated data backup and point-in-time recovery mechanisms
5. Develop a web-based frontend dashboard for management reporting
6. Integrate a payment gateway (e.g., Telebirr, CBE Birr) for digital payment processing

## References

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## Annex A: Survey / Interview Questions

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**Project Title:** Hotel Management System

**Organization:** Addis Grand Hotel, Addis Ababa

**Interview Date:** April 2026 | **Conducted by:** Group 7

### Section 1: Current System & Challenges

7. How do you currently manage room reservations and bookings?
8. What are the biggest problems you face with the existing manual or spreadsheetbased system?
9. How do you handle guest check-in and check-out processes?
10. How do you track room availability and room status throughout the day?
11. What challenges do you encounter with billing and payment record-keeping?

### Section 2: Guest Management

12. How do you currently store guest information (name, contact details, ID number, etc.)?
13. Do guests frequently make special requests? If so, how are these recorded and fulfilled?
14. How do you currently handle guest complaints or feedback after their stay?

### Section 3: Room Service & Additional Services

15. How do you manage room service orders and restaurant orders placed by in-house guests?
16. How do you calculate and record payments for additional services beyond room charges?

### Section 4: Reporting & Future Needs

17. What types of reports do you require on a daily or weekly basis (e.g., occupancy rate, revenue summary, guest list)?
18. What specific improvements would you like to see in a new digital management system?

## Annex B: Sample Forms

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### Form 1: Guest Registration

**Guest Full Name:** \_\_\_\_\_

Phone Number: \_\_\_\_\_

Email Address: \_\_\_\_\_

Nationality: \_\_\_\_\_

ID / Passport Number: \_\_\_\_\_

Home Address: \_\_\_\_\_

### Form 2: Room Booking

Guest Name: \_\_\_\_\_

Check-in Date: \_\_ / \_\_ / \_\_\_\_

Check-out Date: \_\_ / \_\_ / \_\_\_\_

Room Type: ☐ Single ☐ Double ☐ Deluxe ☐ Suite

Number of Guests: \_\_\_\_\_

Special Requests: \_\_\_\_\_

### Form 3: Check-in / Check-out

Booking ID: \_\_\_\_\_

Guest Name: \_\_\_\_\_

Room Number: \_\_\_\_\_

Check-in Date & Time: \_\_\_\_ / \_\_\_\_ / \_\_\_\_ : \_\_\_\_

Check-out Date & Time: \_\_\_\_ / \_\_\_\_ / \_\_\_\_ : \_\_\_\_

### Form 4: Room Service Order

Room Number: \_\_\_\_\_

Guest Name: \_\_\_\_\_

Order Date: \_\_ / \_\_ / \_\_\_\_

| Item Description    | Qty | Unit Price (ETB) | Total (ETB) |
|---------------------|-----|------------------|-------------|
|                     |     |                  |             |
|                     |     |                  |             |
|                     |     |                  |             |
|                     |     |                  |             |
| TOTAL AMOUNT (ETB): |     |                  |             |

Special Instructions: \_\_\_\_\_

### Form 5: Payment Receipt

**Receipt No.:** \_\_\_\_\_

**Guest Name:** \_\_\_\_\_

**Booking ID:** \_\_\_\_\_

**Amount Paid (ETB):** \_\_\_\_\_

**Payment Method:** ☐ Cash   ☐ Debit/Credit Card   ☐ Mobile Money (Telebirr / CBE Birr)

**Date:** \_\_ / \_\_ / \_\_\_\_\_

**Received by:** \_\_\_\_\_

*Thank you for choosing Addis Grand Hotel*

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